Task : SOC Alert Management and Incident Response Operations

1. Objective

The objective of this task is to develop a comprehensive understanding of SOC alert management and incident response operations by studying theoretical concepts such as alert prioritization, incident classification, and incident response lifecycle.
The task also focuses on practical implementation of SOC workflows including alert triage, incident documentation, evidence preservation, and full attack-to-response simulation using security tools in a controlled lab environment.
Through this task, the goal is to build hands-on skills required for a SOC analyst role, including decision-making, threat analysis, and effective incident handling.

2. Alert Priority Levels

2.1 Overview

Alert prioritization is a critical function in a Security Operations Center (SOC) as it helps analysts determine which security alerts require immediate attention. Since organizations generate thousands of alerts daily, proper prioritization ensures that high-risk threats are handled quickly while minimizing alert fatigue.

2.2 Priority Levels Definition

Alerts are categorized into different severity levels based on their impact and urgency:

- **Critical:** Represents severe threats that can cause major damage such as ransomware attacks, data breaches, or system compromise. These require immediate action.
- **High:** Indicates serious incidents like unauthorized administrative access or exploitation of known vulnerabilities.
- **Medium:** Includes suspicious activities such as repeated failed login attempts or abnormal user behavior.
- **Low:** Represents minor events like port scans or informational alerts that require monitoring but not immediate action.

2.3 Alert Prioritization Criteria

Alert severity is determined using multiple factors:

- **Asset Criticality:** Systems like production servers or databases are given higher priority compared to test environments.
- **Exploit Likelihood:** Vulnerabilities with publicly available exploits (e.g., known CVEs) are considered higher risk.
- **Business Impact:** Incidents affecting sensitive data or financial systems are prioritized higher.
- **Threat Context:** Alerts linked with known threat actors or ongoing attacks are treated with higher urgency.

2.4 Scoring Systems

The **Common Vulnerability Scoring System (CVSS)** is widely used to quantify risk:

- Scores range from **0 to 10**
- Example:
  - 9.0 – 10.0 → Critical
  - 7.0 – 8.9 → High
  - 4.0 – 6.9 → Medium

For example, the Log4Shell vulnerability (CVE-2021-44228) has a CVSS score of 9.8, making it a Critical threat.

In addition, modern SIEM tools such as Splunk use **risk-based scoring models** to dynamically prioritize alerts.

2.5 Practical Considerations in Alert Prioritization

Beyond standard scoring, advanced SOC environments also consider:

- **False Positive Rate:** Frequent false alerts are deprioritized over time
- **Threat Intelligence Feeds:** Integration with external intelligence sources
- **User Behavior Analytics (UBA):** Detecting anomalies in user actions
- **Attack Chain Stage:** Early-stage vs. late-stage attack detection

2.6 Key Objective

Two The main objective is to develop the ability to quickly evaluate alerts using structured criteria and ensure efficient incident handling.

3. Incident Classification

3.1 Overview

Security monitoring aims to achieve the following:

- Detect anomalies in network traffic and user behavior patterns
- Identify unauthorized access attempts to systems and data
- Find policy violations by users or applications

3.2 Incident Categories

Security incidents are broadly classified into:

- **Malware Attacks:** Viruses, ransomware, trojans
- **Phishing Attacks:** Fraudulent emails or links
- **Denial of Service (DoS/DDoS):** Service disruption attacks
- **Insider Threats:** Malicious or negligent employees
- **Data Exfiltration:** Unauthorized data transfer
- **Brute Force Attacks:** Password guessing attempts

3.3 Classification Frameworks

Standard frameworks are used for consistency:

- MITRE ATT&CK
  Maps attacker techniques such as:
  - T1566 → Phishing
  - T1190 → Exploit Public-Facing Application
- ENISA Incident Taxonomy
  Provides structured classification categories.
- VERIS Framework
  Focuses on event recording and incident sharing.

3.4 Contextual Metadata

Each incident is enriched with additional details:

- Source and destination IP addresses
- Timestamps of activity
- Affected systems
- Indicators of Compromise (IOCs) such as:
    - File hashes
    - Malicious URLs
    - Suspicious domains

3.5 Extended Classification Factors

To improve analysis, SOC analysts also include:

- **Kill Chain Mapping:** Identifying attack stage (Reconnaissance, Exploitation, etc.)
- **Threat Actor Attribution:** Linking incidents to known attacker groups
- **Geolocation Data:** Identifying origin of attack traffic
- **Behavioral Indicators:** Patterns such as lateral movement

3.6 Key Objective

The objective is to accurately classify and enrich incidents to improve investigation speed and decision-making.

4. Basic Incident Response

4.1 Overview

Incident response is a structured approach used to detect, contain, and recover from cybersecurity incidents. It ensures minimal damage and quick restoration of services.

4.2 Incident Response Lifecycle

Different The incident response process follows the framework defined by NIST SP 800-61:

1. **Preparation:** Setting up tools, policies, and response plans.

2. **Identification:** Detecting and confirming security incidents.
3. **Containment:** Limiting the spread of the attack (e.g., isolating systems).
4. **Eradication:** Removing malware or malicious components.
5. **Recovery:** Restoring systems and services to normal operations.
6. **Lessons Learned:** Documenting findings and improving future responses.

4.3 Incident Response Procedures

Common response actions include:

- System isolation from network
- Blocking malicious IP addresses
- Collecting evidence such as logs and memory dumps
- File hashing for integrity verification
- Communication with stakeholders and management

4.4 Automation and SOAR

Modern SOCs use automation tools like Splunk Phantom to:

- Automate repetitive tasks
- Trigger predefined workflows
- Reduce response time

4.5 Evidence Preservation

Maintaining proper evidence is critical for investigation:

- Use of forensic tools to capture volatile data
- Maintaining chain of custody
- Hashing files using SHA-256 to ensure integrity

4.6 Best Practices in Incident Response

- Maintain incident response playbooks
- Conduct regular tabletop exercises
- Ensure proper documentation of every action
- Follow least privilege principle during response

4.7 Key Objective

The objective is to effectively handle incidents using a structured approach while minimizing damage and ensuring quick recovery.

5. Alert Management Practice

5.1 Overview

Alert management is a critical function in a Security Operations Center (SOC) that involves classification, prioritization, and response to security alerts. This practical task focuses on simulating alert handling processes using structured datasets and documentation.

Since the deployment of a full SIEM solution was not feasible in the current environment, alert management activities were performed using spreadsheet-based classification and manual documentation techniques. This approach still reflects real-world SOC workflows where analysts analyze, prioritize, and respond to alerts efficiently.

5.2 Alert Classification System

An alert classification system was created using Google Sheets to organize alerts based on type, priority, and MITRE ATT&CK techniques.

The following alert categories were included:

Alert ID	Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	SSH Brute Force	Medium	T1110
003	Exploit Attempt	Critical	T1190

Figure 1: Alert classification table

5.3 Alert Prioritization

Alerts were prioritized based on severity, impact, and exploitability using CVSS scoring concepts.
Examples:

- Log4Shell vulnerability (CVSS 9.8) → Critical
- SSH brute-force attempts → Medium
- Port scanning activity → Low

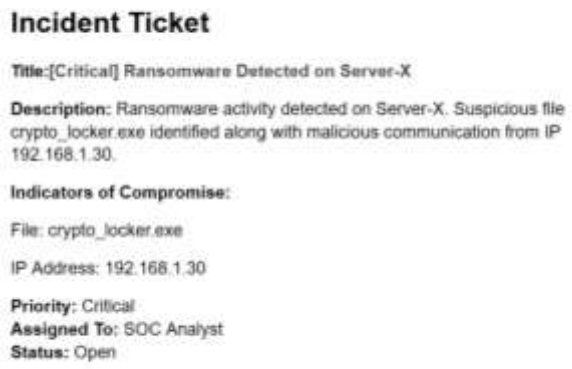
The prioritization process ensures that high-risk threats are addressed first while minimizing unnecessary effort on low-impact alerts.

5.4 Simulated Dashboard Concept

An incident ticket was created to simulate SOC incident tracking and management. The ticket includes key details such as description, indicators of compromise, and priority level.
This helps in maintaining proper documentation and accountability during incident handling.

5.5 Incident Ticket Creation

Although a live dashboard could not be implemented, alert distribution was conceptually visualized based on priority levels. Critical alerts were considered highest in number during active attack simulation, followed by medium and low severity alerts.
This demonstrates how SOC dashboards provide quick visibility into the overall security posture.



Incident Ticket

Title:[Critical] Ransomware Detected on Server-X

Description: Ransomware activity detected on Server-X. Suspicious file crypto_locker.exe identified along with malicious communication from IP 192.168.1.30.

Indicators of Compromise:

File: crypto_locker.exe

IP Address: 192.168.1.30

Priority: Critical
Assigned To: SOC Analyst
Status: Open

Figure 2: Incident ticket creation

5.6 Response Procedure Documentation

A structured response procedure was documented outlining steps to handle alerts. This includes identification, validation, containment, and reporting actions.
Such procedures are essential in ensuring consistency and efficiency in SOC operations.

Alert Response Procedure

Alert Handling and Response Procedure

1. Identify alert source and type
2. Validate alert authenticity
3. Check related logs and indicators
4. Determine severity and priority
5. Isolate affected system if required
6. Document findings
7. Escalate if critical

Figure 3: Documentation Procedure

5.7 Escalation Process

A simulated escalation email was drafted to represent communication with higher-level analysts. Critical alerts require immediate escalation to Tier 2 teams for deeper investigation. This highlights the importance of communication in incident response workflows.

Escalation Email

Subject: Escalation: Critical Ransomware Alert

Dear Tier 2 Team,

A critical ransomware incident has been detected on Server-X. The system shows signs of malicious encryption activity linked to the file crypto_locker.exe. Network logs indicate suspicious communication from IP address 192.168.1.30.

Initial containment steps have been taken, including isolating the affected system from the network. Further analysis is required to assess the extent of the compromise and ensure complete remediation.

Please prioritize this incident for immediate investigation and response.

Regards,
SOC Analyst

Figure 4: Escalation Email Draft

5.8 Considerations in Alert Management

In addition to basic classification and prioritization, advanced SOC environments consider:

- Alert correlation across multiple sources
- Integration with threat intelligence platforms
- Reduction of false positives through tuning
- Risk-based alert scoring models

These factors improve the efficiency and accuracy of alert handling.

5.9 Key Objective

The objective of this practical task was to simulate alert management operations, including classification, prioritization, documentation, and escalation, to develop a clear understanding of SOC workflows.

6. Response Documentation

6.1 Overview

Response documentation is a crucial aspect of incident handling in a Security Operations Center (SOC). It ensures that all actions taken during an incident are properly recorded, enabling effective analysis, accountability, and future improvements.

In this task, incident response activities were simulated using structured documentation. Standard templates were created to represent real-world reporting practices, including incident summaries, timelines, investigation steps, and post-incident analysis.

6.2 Incident Response Template

An incident response template was developed using Google Docs to document a simulated phishing attack scenario. The template follows industry practices inspired by SANS methodology.

The report includes key sections such as executive summary, timeline, impact analysis, remediation steps, and lessons learned.

This structured approach ensures that all critical aspects of the incident are captured systematically.

Incident Response Report

Incident Response Report – Phishing Attack Simulation

1. **Executive Summary**
A simulated phishing attack was conducted to evaluate the incident response process. The attack involved a deceptive email attempting to trick users into clicking a malicious link. The incident was identified and handled using structured response procedures.

2. **Timeline**

Time	Action
14:00	Phishing email received
14:10	User reported suspicious activity
14:20	System isolated
14:40	Analysis completed

3. **Impact Analysis**
No actual compromise occurred, as the scenario was simulated. However, the incident demonstrated potential risks such as credential theft and unauthorized access if not handled properly.

4. **Remediation Steps**

- Blocked malicious email source
- Verified system integrity
- Educated users about phishing risks
- Improved email filtering mechanisms

5. **Lessons Learned**
Early detection and quick reporting significantly reduce the risk of successful phishing attacks. User awareness and proper response procedures play a key role in minimizing impact.

Figure 5: Phishing Incident template report

6.3 Investigation Documentation

All investigation steps were recorded in a structured format with timestamps to track actions performed during the incident.

Timestamp	Action
2025-08-18 14:00:00	Endpoint isolated
2025-08-18 14:30:00	Memory data considered for analysis

This logging approach helps maintain transparency and provides a clear audit trail of incident response activities.

Incident Investigation Log

Timestamp	Action
2025-08-18 14:00:00	Endpoint isolated
2025-08-18 14:30:00	Suspicious activity reviewed
2025-08-18 15:00:00	Indicators analyzed

Figure 6: Incident investigation log

6.4 Checklist Creation

A phishing investigation checklist was created to standardize the analysis process. The checklist ensures that no critical verification step is missed during incident handling.
Key checklist items include verification of email headers, validation of suspicious links using threat intelligence platforms, and identification of affected users.

Phishing Investigation Checklist

1	Verify email headers	
2	Analyze sender domain	
3	Check link reputation using VirusTotal	
4	Identify affected users	
5	Report findings	

Figure 7: Investigation checklist of Phishing

6.5 Post-Incident Analysis

A post-mortem analysis was conducted to evaluate the response process and identify areas for improvement. This step is essential for strengthening future incident response capabilities.
The analysis focused on improving detection speed, enhancing user awareness, and refining response procedures.

6.6 Documentation Practices

In addition to basic documentation, advanced SOC environments incorporate:

- Automated incident reporting systems
- Integration with case management platforms
- Detailed forensic documentation
- Compliance and audit reporting

These practices enhance efficiency and support organizational security governance.

6.7 POST-MORTEM

The simulated phishing incident highlighted the importance of early detection and structured response procedures. The response process was effective in identifying and isolating the threat.

Improvements can be made in user awareness and faster reporting to reduce response time and minimize potential risks in real-world scenarios.

6.8 Key Objective

The objective of this task was to develop structured documentation skills for incident response, ensuring clear communication, effective tracking, and continuous improvement in handling security incidents.

7. Alert Triage Practice

7.1 Overview

Alert triage is a critical process in a Security Operations Center (SOC) that involves analyzing alerts to determine their validity, severity, and required response actions. It helps in distinguishing between true positives, false positives, and benign activities.
In this task, alert triage was simulated using controlled attack scenarios and threat intelligence validation techniques. Since a live SIEM environment was not available, alerts were generated and analyzed manually to replicate real-world triage workflows.

7.2 Triage Simulation

A simulated alert representing a brute-force SSH attack was analyzed. The alert was structured to include key details such as description, source IP, priority, and status.

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	192.168.65.129	Medium	Open

The alert indicates repeated login attempts from a single source, which is a common sign of brute-force activity. Based on its behavior and impact, it was categorized as a medium-priority alert.

Figure 8: Brute force attack using Hydra

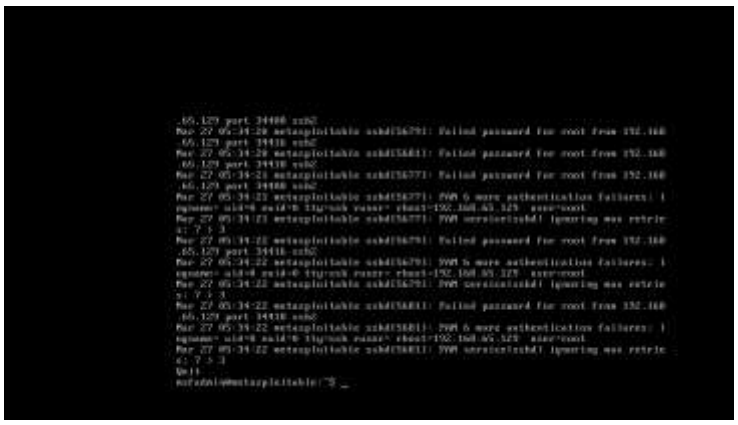


Figure 9: Failed brute force attempts

7.3 Alert Analysis

The simulated brute-force activity was generated in a controlled environment using penetration testing techniques. The repeated authentication attempts were observed and analyzed to understand attack patterns.

Such alerts require verification to determine whether they represent malicious activity or normal user behavior. In real-world scenarios, analysts correlate logs and historical data before making a decision.

7.4 Threat Intelligence Validation

To validate the alert, the source IP was checked using external threat intelligence platforms such as VirusTotal and AlienVault OTX. The lookup helps determine whether the IP address has been previously associated with malicious activities.

Since the activity was performed in a controlled lab environment, the IP did not show malicious reputation, confirming it as a simulated scenario rather than a real threat.

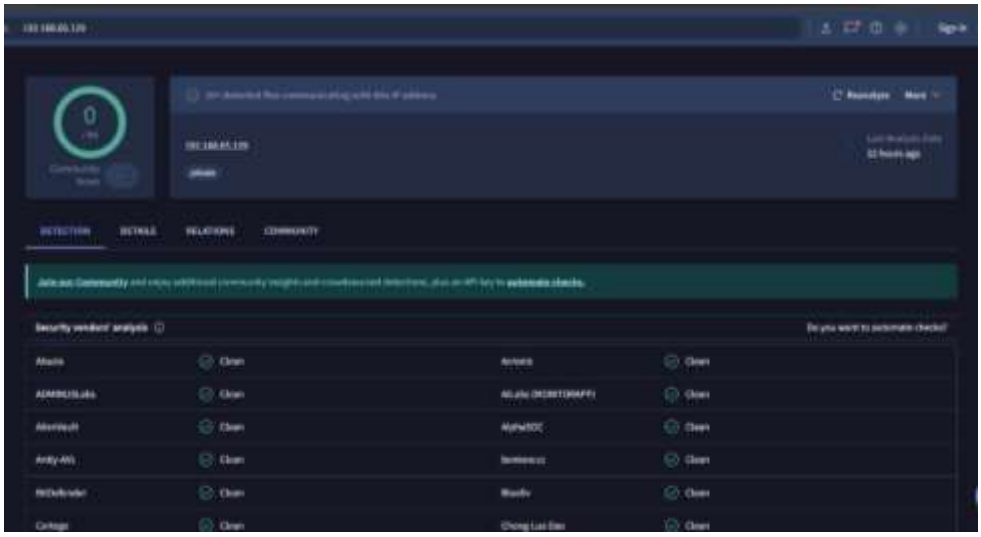


Figure 10: Threat Validation on VirusTotal

7.5 False Positive Consideration

An important aspect of triage is identifying false positives. Alerts may sometimes be triggered by legitimate activities such as system scans or administrative actions.

In this case, the alert was verified and classified as a controlled test activity, demonstrating how analysts differentiate between actual threats and expected behavior.

7.6 Triage Techniques

Modern SOC environments enhance triage using:

- Behavioral analysis
- Log correlation across systems
- Automated alert enrichment
- Integration with threat intelligence feeds

These techniques improve accuracy and reduce analyst workload.

7.7 Findings

The brute-force attack simulation generated repeated login attempts, triggering a security alert. The alert was analyzed and categorized as medium priority. Threat intelligence validation showed no malicious reputation for the source IP, confirming it as a controlled test. This demonstrates the importance of verifying alerts before taking action.

Alert Triage Report – SSH Brute Force

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	192.168.65.129	Medium	Open

Analysis: The alert represents multiple failed login attempts from a single source IP. This behavior is typical of brute-force attacks targeting SSH services. Based on the frequency and pattern, the alert was categorized as medium priority.

Threat Intelligence Validation:
The source IP was analyzed using VirusTotal and AlienVault OTX. No malicious activity was associated with the IP, indicating that the alert was generated as part of a controlled testing environment.

Figure 11: Alert Triage report

7.8 Key Objective

The objective of this task was to simulate the alert triage process, analyze security alerts, validate indicators of compromise, and understand the importance of distinguishing between true and false positives.

8. Evidence Preservation

8.1 Overview

Evidence preservation is a critical component of incident response that ensures collected data remains intact, reliable, and admissible for analysis or investigation. It involves capturing system data, maintaining integrity through hashing, and documenting the chain of custody.
In this task, evidence preservation techniques were simulated using system-level commands and existing tools. Due to system constraints, lightweight methods were used to replicate the process of collecting and documenting digital evidence.

8.2 Volatile Data Collection

Volatile data refers to information that is lost when a system is powered off, such as active network connections and running processes. This data was collected using system commands to simulate real-time evidence acquisition.

The netstat command was used to capture active network connections, which helps in identifying suspicious communication between systems.

```
┌──(kali㉿kali)-[~]
└─$ sudo netstat -antp
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
```

Figure 12: Netstat active network connection capture

8.3 Evidence Collection and Integrity

A sample file was created and its hash value was generated using SHA-256 to ensure data integrity. Hashing ensures that the evidence has not been altered after collection.
Maintaining integrity is essential in forensic investigations, as even minor modifications can invalidate the evidence.

```
┌──(kali㉿kali)-[~]
└─$ sha256sum evidence.txt
3841b155f70ee881690bf89dab7a62fa7361327441b437225abd07a7313933b8  evidence.txt
```

Figure 13: SHA-256 hash generation

8.4 Chain of Custody Documentation

A structured table was created to document collected evidence, including details such as description, date, and hash value. This ensures proper tracking and accountability of evidence throughout the investigation process.

Evidence Collection and Preservation Report

Item	Description	Collected By	Date	Hash Value
Network Data	Active connections captured using netstat	SOC Analyst	Today	-
File Evidence	Sample file evidence.txt	SOC Analyst	Today	3841b155f70ee881690bf89dab7a62fa7361327441b437225abd07a7313933b8

Evidence Handling Notes: The evidence was collected using system-level commands and preserved by generating a SHA-256 hash. This ensures that the integrity of the data remains intact. Proper documentation was maintained to track the evidence throughout the process.

Figure 14: Evidence Collection Report

8.5 Evidence Handling

In real-world scenarios, advanced tools such as Velociraptor and FTK Imager are used for memory acquisition and forensic imaging. These tools provide deeper visibility into system memory and disk data.

Additionally, proper evidence handling includes secure storage, access control, and maintaining logs of all actions performed on the evidence.

8.6 Key Objective

The objective of this task was to understand the importance of evidence preservation, practice basic data collection techniques, and document evidence in a structured and reliable manner.

9. Capstone Project: Full Alert-to-Response Cycle

9.1 Overview

The capstone project simulates a complete security incident lifecycle, including attack execution, detection, triage, response, and reporting. This provides a comprehensive understanding of how Security Operations Centers (SOC) handle real-world cyber threats.

In this task, a controlled attack was performed using a vulnerable system, and the full response workflow was documented. Due to environmental limitations, detection and response actions were simulated using structured analysis and documentation techniques.

9.2 Attack Simulation

A vulnerability in the Metasploitable system was exploited using Metasploit. The VSFTPD backdoor exploit was used to gain unauthorized access to the target system.

This simulation demonstrates how attackers exploit known vulnerabilities to compromise systems.

Figure 15: Root Access granted & shell working

9.3 Detection and Triage

The attack activity was documented and analyzed to simulate detection. The following alert was generated:

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-27 11:00:00	192.168.65.129	VSFTPD Exploit Attempt	T1190

The alert was categorized and analyzed to understand the nature and impact of the attack.

9.4 Incident Response

Upon identifying the attack, response actions were simulated. The affected system was logically isolated, and further communication from the attacker was considered blocked.
This step represents containment strategies used to prevent further compromise.

9.5 Reporting and Documentation

A detailed incident report was created using Google Docs. This ensures proper documentation and communication of the incident.

Incident Report – VSFTPD Exploit Simulation

A controlled cyber attack simulation was conducted to assess the incident response process. The attack targeted a vulnerable FTP service on the Metasploitable system using a known exploit.

The attacker successfully gained unauthorized access by exploiting the VSFTPD backdoor vulnerability. This demonstrated how unpatched services can be compromised.

The activity was identified and analyzed as a high-severity alert. The affected system was logically isolated to prevent further unauthorized access.

The incident response process included identifying the attack vector, analyzing the impact, and documenting the findings. Since the environment was controlled, no actual damage occurred.

This simulation highlights the importance of timely patching, continuous monitoring, and effective incident response procedures. It also demonstrates how attackers can exploit known vulnerabilities if systems are not properly secured.

Figure 16: Incident Report

9.6 Stakeholder Communication

A simplified briefing was prepared for non-technical stakeholders to explain the incident, its impact, and actions taken. This highlights the importance of clear communication during security incidents.

Stakeholder Brief

A simulated cyber attack was conducted to evaluate system security and response capabilities. The attack targeted a known vulnerability in an FTP service and successfully demonstrated how unauthorized access can be gained.

The activity was identified and handled promptly, and necessary containment measures were applied. Since this was a controlled environment, no real impact occurred.

This exercise helped in understanding how attacks happen and how they can be managed effectively. It also emphasized the importance of system updates, monitoring, and quick response to potential threats to maintain overall security.

Figure 17: Stakeholder brief

9.7 Considerations

In real-world SOC environments, such scenarios involve:

- Automated detection through SIEM tools
- Real-time blocking using security platforms
- Continuous monitoring and alert correlation
- Integration with threat intelligence feeds

9.8 Key Objective

The objective of this capstone task was to simulate a full alert-to-response cycle, demonstrating the ability to detect, analyze, respond, and document a security incident effectively.

10. Challenges Faced

Several challenges occurred during this task.

- System limits stopped advanced forensic tools, used basic commands like netstat.
- Lab attacks lacked real network traffic, harder to test false positives.

These issues taught how to adapt in limited settings.

11. Key Learnings

This task gave clear insights into SOC work.

- CVSS scores and criteria help focus on high-risk alerts first.
- NIST steps and checklists make response consistent and fast.
- Hashing and data capture keep evidence valid for investigations.
- Full simulations from alert to report build real decision skills.
- Manual methods work when tools are missing.

The mix of theory and practice prepares for SOC analyst jobs.

12. Conclusion

This task covered SOC alert management and incident response well. It included theory and practice for skills like triage, classification, and full response cycles.

Simulations used Metasploit, Hydra, and VirusTotal to show real threats despite lab limits. NIST, MITRE, and CVSS standards were followed.

The work shows SOC analysts must prioritize, document, and keep learning to protect organizations.